## Chapter 1

# Real Numbers

History of numbers is as old as that of human civilization. Mathematics started with expressing quantity using numbers. According to Greek Philosopher Aristotle formal beginning of mathematics started with its practice among ancient Egyptian priests. So it can be said that number based Mathematics was created some two thousand years before the birth of Jesus Christ. Numbers and operations on them have reached today's universal form after that through the hands of different nations and civilizations.

Indian mathematicians introduced for the first time the concept of 0 and decimal number system for satisfying the need of counting natural numbers. This is considered as a milestone in the representation of numbers. Later on Indian and Chinese mathematicians expanded the concepts of 0, negative, real, integer and fractional numbers which were taken as the basis by Arabian mathematicians of medieval age. Muslim mathematicians of the middle east are given the credit of expressing numbers using decimal fractions. Again they first introduced irrational numbers in the form of square roots for solution of quadratic algebraic equations in 11th century. Historians think that around 500 A.D. Greek philosophers felt necessity of irrational numbers, especially square root of 2, for the purpose of geometrical drawing. In the nineteenth century European mathematicians gave completeness to real numbers through introducing complete set of rules of operations. In order to satisfying daily needs students are required to have clear understanding about real numbers. In this chapter real numbers have been discussed in general. At the end of this chapter students will learn —

- How to classify real numbers.
- How to deduce approximate values of real numbers by expressing it in decimal fractions.
- How to classify decimal fraction.

- How to interpret repeated decimal numbers and convert fractions into repeated decimal numbers.
- How to convert repeated decimal fractions into fractions.
- How to interpret infinite nonrepeating fractional numbers.
- How to interpret similar and dissimilar decimal fractions.
- How to carry out addition, subtraction, multiplication and division operations on repeated decimal fractions, and will be able to solve different relevant problems.

# Classification of Real Numbers

**Natural Number**: $1, 2, 3, 4, \ldots$ etc are natural numbers. $2, 3, 5, 7, \ldots$ are primes and $4, 6, 8, 9, \ldots$ etc are composite numbers. If GCD(Greatest Common Divisor) of two integers is 1 then they are called mutually primes. For example 6 and 35 are mutually primes since they do not have a common division other than 1.

**Integer**: All positive and negative fractionless numbers are said to be integers. For example, $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ etc are integers.

**Fractional Number**: Numbers expressible as $\frac{p}{q}$ are said to be factional numbers where $q \neq 0$ and $q \neq 1$ For example, $\frac{1}{2}, \frac{3}{2}, \frac{-5}{3}, \frac{4}{6}$ etc are simple fractional numbers. In case of a simple fraction $\frac{p}{q}$ if $p < q$ then the fraction is called proper and if $p > q$ then the fraction is termed improper. For example $\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \ldots$ etc are proper fractions. $\frac{3}{2}, \frac{4}{3}, \frac{5}{3}, \frac{5}{4}, \ldots$ are improper fractions.

**Rational Number**: Numbers of the form $\frac{p}{q}$ are called rational when $p$ and $q$ are integers and $q \neq 0$. For example, $\frac{3}{1} = 3$, $\frac{11}{2} = 5.5$, $\frac{5}{3} = 1.666, \ldots$ etc are rational numbers. Any rational number can be expressed as ratio of two mutually prime numbers. All integers and fractions are rational numbers.

**Irrational Number**: If a number cannot be expressed in the form $\frac{p}{q}$ where $p$ and $q$ are integers and $q \neq 0$, then it is called an irrational number. Root of a natural number not equal to perfect square or its fraction is an irrational number.

For example, $\sqrt{2} = 1.414213\ldots$, $\sqrt{3} = 1.732\ldots$, $\frac{\sqrt{5}}{2} = 1.118\ldots$, are irrational numbers. No irrational number can be expressed as ratio of two integers.

**Decimal Fractional Number**: Rational and irrational numbers when expressed using decimal sign is called decimal fractional number. For example, $3 = 3.0$, $\frac{5}{2} = 2.5$, $\frac{10}{3} = 3.3333\ldots$, $\sqrt{3} = 1.732\ldots$, etc are decimal fractional numbers. If there are finite number of digits after decimal sign then this is called finite decimal fraction and if there are infinitely many digits then it is called infinite decimal fractions. For example, 0.52, 3.4152 etc are finite decimal fractions, whereas $\frac{4}{3} = 1.333\ldots, \sqrt{5} = 2.123512367\ldots$ etc are infinite decimal fractions. Again, in case of infinite decimal numbers if some degits after decimal point are repeated they are called infinite repeated decimal fractions. If digits are not repeated they are called infinite decimal numbers without repeatation. For example, $\frac{122}{99} = 1.2323\ldots, 5.1\dot{6}5\dot{4}$ etc are infinite repeated decimal fractions and $0.523050056\ldots, 2.12340314\ldots$ etc are infinite decimal numbers without repeatation.

**Real Number**: All rational and irrational numbers are called real numbers. For example, the following numbers are real numbers.

$0, \pm 1, \pm 2, \pm 3, \cdots$ $\pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{4}{3}, \cdots$

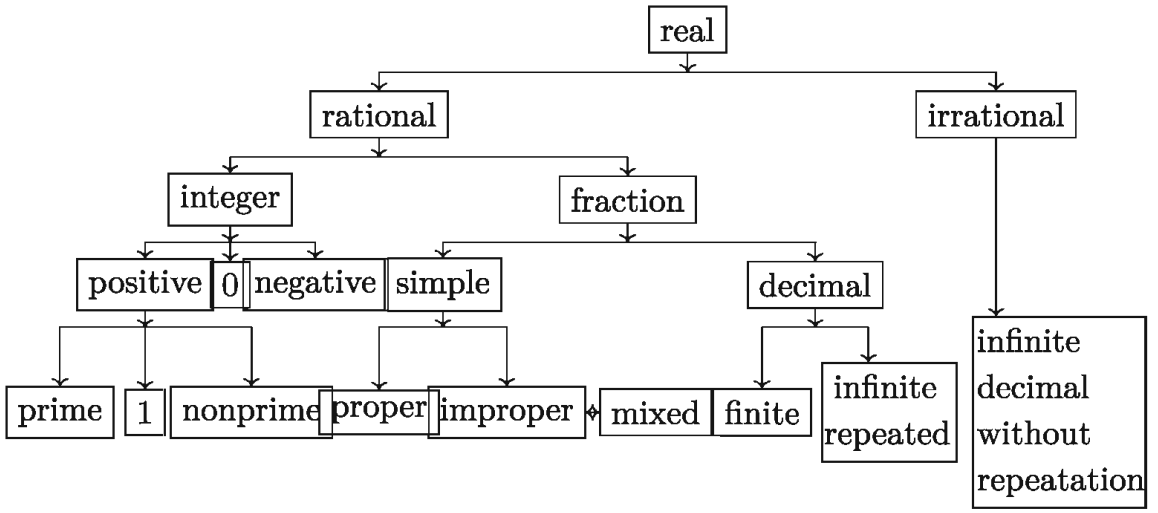
$\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \cdots$ $1.23, 0.415, 1.3333\ldots, 0.6\dot{2}\dot{2}, 4.120345061\ldots$

**Positive Number**: Numbers bigger than 0 are called positive numbers. For example, $2, \frac{1}{2}, \frac{3}{2}, \sqrt{2}, 0.415, 0.6\dot{2}\dot{2}, 4.120345061\ldots$, etc are positive numbers.

**Negative Number**: Numbers smaller than 0 are called negative numbers. For example, $-2, -\frac{1}{2}, -\frac{3}{2}, -\sqrt{2}, -0.415, -0.6\dot{2}\dot{2}, -4.120345061\ldots$, etc are negative numbers.

**Non-negative Number**: Positive numbers including 0 are called nonnegative numbers. For example, $0, 3, \frac{1}{2}, 0.612, 1.\dot{3}, 2.120345\ldots$, etc are nonnegative numbers.

In the following chart we can see classification of real numbers.




**Work:** Show the position of the following numbers in the classification of real numbers $\frac{3}{4}$, 5, $-7$, $\sqrt{13}$, 0, 1, $\frac{9}{7}$, 12, $2\frac{4}{5}$, 1.1234, $0.3\dot{2}\dot{3}$

**Example 1.** Find an irrational number between $\sqrt{3}$ and 4.

**Solution:** Here, $\sqrt{3} = 1.7320508\ldots$

Suppose, $a = \dfrac{\sqrt{3}+4}{2} \approx 2.866$ and $b = \dfrac{\sqrt{3}+4+4}{3} \approx 3.244$

Obviously $a$ and $b$ are both real numbers and both are bigger than $\sqrt{3}$ and smaller than 4 because $a$ is average of two uneqal numbers $\sqrt{3}$ and 4, and $b$ is average of numbers $\sqrt{3}$, 4 and 4.

That is, $\sqrt{3} < 2.866 < 4$ and $\sqrt{3} < 3.244 < 4$

Again, $a$ and $b$ cannot be expressed as fractions.

$\therefore$ $a$ and $b$ are the desired numbers.

In fact we can construct countless such irrational numbers.

**Basic characteristics of addition and multiplication processes of real numbers** :

1. If $a$ and $b$ are real numbers $(i)$ $a+b$ is a real number and $(ii)$ $ab$ a real number.
2. If $a$ and $b$ are real numbers $(i)$ $a+b=b+a$ and $(ii)$ $ab=ba$
3. If $a,\ b,\ c$ are real numbers $(i)$ $(a+b)+c=a+(b+c)$ and $(ii)$ $(ab)c=a(bc)$
4. If $a$ is real number, there are only two real numbers 0 and 1 such that $(i)$ $0 \neq 1$, $(ii)$ $a+0=0+a=a$ and$(iii)$ $a\cdot 1=1\cdot a=a$

5. If $a$ is real number, $(i)\ a+(-a)=0$ $(ii)$ If $a \neq 0$ then $a \cdot \frac{1}{a}=1$
6. If $a, b, c$ are real numbers, $a(b+c)=ab+ac$
7. If $a, b$ are real numbers $a<b$ or $a=b$ or $a>b$
8. If $a, b, c$ are real numbers and $a<b$ then $a+c<b+c$
9. If $a, b, c$ are real numbers and $a<b$ then $(i)\ ac<bc$ whenever $c>0$ $(ii)\ ac>bc$ whenever $c<0$

**Proposition:** $\sqrt{2}$ is an irrational number.

**Proof:** Suppose $\sqrt{2}$ is an irrational number.

Then there are two mutually prime numbers $p,\ q>1$ so that $\sqrt{2}=\frac{p}{q}$

Or, $2=\frac{p^2}{q^2}$ [squaring] That is $2q=\frac{p^2}{q}$ [Multiplying both sides by $q$ ]

Obviously $2q$ is an integer but $\frac{p^2}{q}$ is not an integer, because $p$ and $q$ are natural numbers and they are mutually prime and $q>1$.

$\therefore 2q$ and $\frac{p^2}{q}$ cannot be equal. That is $2q \neq \frac{p^2}{q}$.

$\therefore \sqrt{2}$ cannot be expressed in the form $\frac{p}{q}$. That is $\sqrt{2} \neq \frac{p}{q}$

$\therefore \sqrt{2}$ is an irrational number. □

**Work:** Prove that $\sqrt{3}$ is an irrational number.

**Example 2.** Prove that when one is added to the product of 4 consecutive natural numbers this is a perfect square.

**Solution:** Let the consecutive integers be $x,\ x+1,\ x+2,\ x+3$

If we add 1 to the product of these 4 integers we get:

$$x(x+1)(x+2)(x+3)+1$$
$$=x(x+3)(x+1)(x+2)+1$$
$$=(x^2+3x)(x^2+3x+2)+1$$
$$=a(a+2)+1 \text{ [Assume } x^2+3x=a \text{ ]}$$
$$=a^2+2a+1=(a+1)^2 \qquad =(x^2+3x+1)^2$$

which is a perfect square. Therefore when 1 is added to the product of any four consecutive natural numbers this is a perfect square.

# Decimal Fractions

Each real number can be expressed in decimal fractions. For example, $2 = 2.0$, $\frac{2}{5} = 0.4$, $\frac{1}{3} = 0.333\ldots$ etc. Decimal fractions are of three types, namely: finite, repeated and infinite decimal fractions.

**Finite decimal fractions**: In any finite decimal fraction there is only a finite numbers of digits after the decimal point. For example, 0.12, 1.023, 7.832, 54.67, ... etc are finite decimal fractions.

**Repeated decimal fractions**: In any repeating decimal fraction all or some digits on the right of decimal point come again and again. For example, $3.333\ldots$, $2.454545\ldots$, $5.12765765\ldots$ etc are repeated decimal fractions.

**Infinite decimal fractions**: In an infinite decimal fraction digits on the right of decimal point or its part will not appear repeatedly. For example, $1.4142135\ldots$, $2.8284271\ldots$ etc are Infinite decimal fractions.

**Remarks:** Finite and repeating decimal fractions are rational numbers whereas infinite decimal numbers are irrational numbers. Value of any irrational number can be computed to any desired decimal place. If numerator and denominator of a fraction can be represented by natural numbers then the fraction is a rational number.

**Work:** Classify the following numbers with reasons
$1.723$, $5.2333\ldots$, $0.0025$, $2.1356124\ldots$, $0.01050105\ldots$ and $0.450123\ldots$

**Repeating decimal fractions**

```
6)23(3.833
  18
  --
   50
   48
   --
    20
    18
    --
     20
     18
     --
      2
```

Express the fraction $\frac{23}{6}$ into decimal fraction. Observe that when numerator was divided by denominator the division process could not be completed. Observe that in the result the digit 3 appears over and over again. So $3.8333\ldots$ is a repeating decimal number.

The decimal fraction for which the same digit or a set of digits in order appear after decimal point is called repeating decimal fraction. The part of the repeating decimal fraction that appears over and over again is called repeating. The remaining part is nonrepating.

In a repeating decimal fraction if a single digit repeats then a point is marked above that digit. Otherwise point is marked on the first and the last repeating digits. For example, $2.555\ldots$ is written as $2.5\dot{}$ and $3.124124124\ldots$ as $3.\dot{1}2\dot{4}$.

In a decimal number if every digit after decimal point is repeating then it is called a pure repeating fraction. Otherwise the fraction is called mixed. For example, $1.\dot{3}$ is a pure repeating fraction and $4.2351\dot{1}\dot{2}$ is a mixed repeating fraction.

If in the denominator there are digits other than 2, 5 as a factor then no numerator will be divisible by that denominator. Since there cannot be any digit other than 1, 2, ..., 9 at some stage as residue the same numbers will appear. Number of digits in the repeating part is always less than the number in the denominator.

**Example 3.** Express $\frac{3}{11}$ and $\frac{95}{37}$ into decimal fractions.

**Solution:** $\frac{3}{11}$ and $\frac{95}{37}$ have been converted into decimal fractions as below.

Actually 3 has been divided. But since 3 is smaller than 11, 0 has been placed in the dividend and a 0 has been placed to the right of 3

```
11)30(0.2727
   22
   80
   77
   30
   22
   80
   77
    3
```

$\therefore \frac{3}{11} = 0.2727\ldots = 0.\dot{2}\dot{7}$

```
37)95(2.56756
   74
   210
   185
    250
    222
     280
     259
      210
      185
       250
       222
        28
```

$\therefore \frac{95}{37} = 2.56756\ldots = 2.\dot{5}6\dot{7}$

Desired decimal fractions are $0.\dot{2}\dot{7}$ and $2.\dot{5}6\dot{7}$

### Converting repeating decimal fractions into common fractions

**Example 4.** Convert the following repeating decimal fractions $0.\dot{3}$, $0.\dot{2}\dot{4}$, and $42.34\dot{7}\dot{8}$ into common fractions.

**Solution:** In the following repeating fractions $0.\dot{3}$, $0.\dot{2}\dot{4}$ and $42.34\dot{7}\dot{8}$ have been converted into common fractions.

$$\begin{aligned} \text{First} \qquad 0.\dot{3} &= 0.3333\ldots \\ 0.\dot{3} \times 10 &= 0.333\ldots \times 10 = 3.333\ldots \\ 0.\dot{3} \times 1 &= 0.333\ldots \times 1 = 0.333\ldots \end{aligned}$$

$$\begin{aligned} \text{subtracting, } 0.\dot{3} \times 10 - 0.\dot{3} \times 1 &= 3 \\ 0.\dot{3} \times (10 - 1) &= 3 \\ 0.\dot{3} \times 9 &= 3 \\ \therefore 0.\dot{3} &= \frac{3}{9} = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{Now} \qquad 0.\dot{2}\dot{4} &= 0.24242424\ldots \\ 0.\dot{2}\dot{4} \times 100 &= 0.242424\ldots \times 100 = 24.24242424\ldots \\ 0.\dot{2}\dot{4} \times 1 &= 0.242424\ldots \times 1 = 0.24242424\ldots \end{aligned}$$

$$\begin{aligned} \text{subtracting, } 0.\dot{2}\dot{4} \times 99 &= 24 \\ \therefore 0.\dot{2}\dot{4} &= \frac{24}{99} = \frac{8}{33} \end{aligned}$$

$$\begin{aligned} \text{Finally} \qquad 42.34\dot{7}\dot{8} &= 42.34787878\ldots \\ 42.34\dot{7}\dot{8} \times 10000 &= 42.34787878\ldots \times 10000 = 423478.78787878\ldots \\ 42.34\dot{7}\dot{8} \times 100 &= 4234.7878\ldots \times 100 = 42.34787878\ldots \end{aligned}$$

$$\begin{aligned} \text{subtracting, } 42.34\dot{7}\dot{8} \times 9900 &= 423478 - 4234 = 419244 \\ \therefore 42.34\dot{7}\dot{8} &= \frac{419244}{9900} = \frac{34937}{825} = 42\frac{287}{825} \end{aligned}$$

$\therefore$ desired common fractions are repsectively $0.\dot{3} = \frac{1}{3}$, $0.\dot{2}\dot{4} = \frac{8}{33}$, $42.34\dot{7}\dot{8} = 42\frac{287}{825}$

**Explanation:** From the three examples above it can be seen that

- Repeating fraction is multiplied with a number equalling 1 followed by as many 0s as there are digits after decimal point of a repeating fraction.

- The repeating decimal fraction has been multiplied with a number equalling 1 followed by as many 0s as many nonrepeating digits are there in the repeating decimal fraction.
- Subtracting the second number from the first results in a whole number. It may be noted that nonrepeating part has been subtracted from the repeating decimal fraction by deleting decimal point and repeating point.
- The result of subtraction has been divided by a number equalling as many 9s as many repeating digits were there in the original repeating decimal fraction, followed by as many 0 as there were nonrepeating digits after decimal point.
- In converting repeating decimal fraction into common fraction we have denominator having as many 9s as many repeating digits followed by as many 0s as many nonrepeating digits after decimal point. Numerator equals a whole number we get after removing decimal and repeating points in the fraction less the number obtained by deleting the decimal point and all digits in the repeating part.

**Remarks:** Repeating decimal fractions can always be converted to common fractions. All repeating decimal fractions are rational numbers.

**Example 5.** Convert $5.23\dot{4}5\dot{7}$ into a common fraction.

**Solution:**

$$\begin{aligned} 5.23\dot{4}5\dot{7} &= 5.23457457457\ldots \\ 5.23\dot{4}5\dot{7} \times 100000 &= 523457.457457\ldots \\ 5.23\dot{4}5\dot{7} \times 100 &= 523.457457\ldots \end{aligned}$$

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$$\text{subtracting, } 5.23\dot{4}5\dot{7} \times 99900 = 522934$$

$$\therefore 5.23\dot{4}5\dot{7} = \frac{522934}{99900} = \frac{261467}{49950} = 5\frac{11717}{49950}$$

$\therefore$ The sought fraction is $5\frac{11717}{49950}$.

**Explanation:** Since there are 5 digits in the decimal part, the repeating decimal fraction has been multiplied by 100000 (5 0s after 1). Since there are two decimal digits before the repeating part 100 (1 followed by 2 0s) repeating decimal fraction has been multiplied by 100. The second product has been deducted from the first. On one side of subtraction is a whole number, and in the other the repeating decimal number multiplied by $(100000 - 1000) = 99900$. Both sides are divided

by 99900 to obtain the common fraction.

**Rules for converting repeating decimal fractions into common fractions:**

Numerator of the desired fraction equals the integer obtained by deleting the decimal point and subtracting nonrepeating portion after the decimal point.

Denominator is obtained by putting as many 9s as many digits in the repeating part followed by as many 0s in the nonrepeating aprt after the decimal point.

These rules have been applied in the following examples to convert a repeating decimal fractions into common fractions.

**Example 6.** Convert $45.2\dot{3}4\dot{6}$ into common fraction.

**Solution:** $45.2\dot{3}4\dot{6} = \dfrac{452346 - 452}{9990} = \dfrac{451894}{9990} = \dfrac{225947}{4995} = 45\dfrac{1172}{4995}$

$\therefore$ the desired fraction is $45\dfrac{1172}{4995}$.

**Example 7.** Convert $32.\dot{5}6\dot{7}$ into common fraction.

**Solution:** $32.\dot{5}6\dot{7} = \dfrac{32567 - 32}{999} = \dfrac{32535}{999} = \dfrac{3615}{111} = \dfrac{1205}{37} = 32\dfrac{21}{37}$

$\therefore$ the desired fraction is $32\dfrac{21}{37}$.

**Work:** Convert $0.\dot{4}\dot{1}$, $3.04\dot{6}2\dot{3}$, $0.0\dot{1}\dot{2}$ and $3.31\dot{2}\dot{4}$ into common fractions.

### Similar and dissimilar repeating decimal fractions

If for two or more repeating decimal fractions number of digits both in repeating and nonrepeating parts are the same then they are called similar repeating decimal fractions. Otherwise they are called dissimilar repeating decimal fractions. For example, $12.\dot{4}\dot{5}$ and $6.\dot{3}\dot{2}$; $9.45\dot{3}$ and $125.89\dot{7}$ are similar repeating decimal fractions. But, $0.3\dot{4}5\dot{6}$ and $7.45\dot{7}8\dot{9}$; $6.43\dot{5}\dot{7}$ and $2.89\dot{3}4\dot{5}$ are dissimilar repeating decimal fractions.

### Converting dissimilar repeating decimal fractions into similar ones

If we write repeating part of a repeating decimal fraction the value does not change. For example $6.45\dot{3}\dot{7} = 6.45\dot{3}73\dot{7} = 6.4537\dot{3} = 6.4537\dot{3}\dot{7}$. Here each one of them is a repeating decimal fraction $6.45373737\ldots$, which is an infinite decimal

number. If we convert this repeating decimal number into common fraction we will see that all of them are equal.

$$6.45\dot{3}\dot{7} = \frac{64537 - 645}{9900} = \frac{63892}{9900}$$
$$6.45\dot{3}73\dot{7} = \frac{6453737 - 645}{999900} = \frac{6453092}{999900} = \frac{63892}{9900}$$
$$6.4537\dot{3}\dot{7} = \frac{6453737 - 64537}{990000} = \frac{6389200}{990000} = \frac{63892}{9900}$$

In order to convert into similar repeating decimal fractions every repeating decimal must have equal number of digits in nonrepeating part. Repeating part must have number of digits equal to lowest common multiple of digits in all repeating parts.

**Example 8.** Convert $5.\dot{6}$, $7.3\dot{4}\dot{5}$, and $10.78\dot{4}2\dot{3}$ into similar repeating decimal fractions.

**Solution:** Repeating decimal fractions $5.\dot{6}$, $7.3\dot{4}\dot{5}$, and $10.78\dot{4}2\dot{3}$ have nonrepeating digits equalling 0, 1 and 2 respectively. Here $10.78\dot{4}2\dot{3}$ has the most number of digits in nonrepeating part, so in order to convert into similar repeating decimal fractions we must have 2 digits in nonrepeating part of every number. $5.\dot{6}$, $7.3\dot{4}\dot{5}$ and $10.78\dot{4}2\dot{3}$ repeating decimal fractions have respectively 1, 2 and 3 digits in the repeating part. LCM of 1, 2 and 3 is 6. So in order to convert into similar repeating decimal fractions we must have 6 digits in the repating part of every number. So, $5.\dot{6} = 5.66\dot{6}666\dot{6}$, $7.3\dot{4}\dot{5} = 7.34\dot{5}45\dot{4}5\dot{4}$ and $10.78\dot{4}2\dot{3} = 10.78\dot{4}23\dot{4}2\dot{3}$. Desired similar repeating decimal fractions are respectively $5.66\dot{6}666\dot{6}$, $7.34\dot{5}45\dot{4}5\dot{4}$ and $10.78\dot{4}23\dot{4}2\dot{3}$.

**Example 9.** Convert $1.7643$, $3.\dot{2}\dot{4}$ and $2.78\dot{3}4\dot{6}$ similar repeating decimal fractions.

**Solution:** In $1.7643$ we have 4 digits in the nonrepeating part and it does not have repeating part. In $3.\dot{2}\dot{4}$ number of digits in nonrepeating part is 0 and that of repeating part is 2; in $2.78\dot{3}4\dot{6}$ number of digits in nonrepeating part is 2 and that in repeating part is 3. Here the largest number of nonrepeating digits is 4 and number of digits in the repeating parts are 2 and 3, LCM of which is 6. So each number must have 4 nonrepating digits and 6 repeating digits.
$\therefore 1.7643 = 1.7643\dot{0}0000\dot{0}$, $3.\dot{2}\dot{4} = 3.2424\dot{2}4242\dot{4}$ $2.78\dot{3}4\dot{6} = 2.7834\dot{6}3463\dot{4}$
Desired similar repeating decimal fractions are respectively $1.7643\dot{0}0000\dot{0}$, $3.2424\dot{2}4242\dot{4}$ and $2.7834\dot{6}3463\dot{4}$

**Remarks:** In order to convert finite decimal fractions into similar decimal fractions we must add sufficient number of 0s at the end of the number to match with number of nonrepeating digits. In case of repeating decimal fractions for each number, numbers of nonrepeating digits have been made equal so should be the case with repating digits.

**Work:** Convert 3.467, $2.01\dot{2}\dot{4}\dot{3}$ and $7.52\dot{5}\dot{6}$ into similar repeating decimal fractions.

### Addition and subtraction of repeating decimal fractions

In order to add repeating decimal fractions we need to convert them into similar repeating decimal fractions. Then addition and subtraction must be carried out as in case of finite decimal fractions. Results thus found will not necessarily be the right results. In order to obtain the correct value we must add or subtract from the last digit of the result, the digit in hand when the leftmost repeating digits were added/subtracted.

**Remark:**

1. Addition and subtraction of repeating decimal fractions result in repeating decimal fractions. In addition and subtration result is yet another repeating decimal fraction having number of nonrepeating digits equalling the largest/larger number of nonrepeating digits among the numbers involved in the operation. Number of digits in the repating part equals LCM of the numbers of digits in the repeating part. In case of finite decimal fractions the number of digits in the nonrepeating part is matched with the largest such number among numbers involved in addition or subtraction by appending appropriate number of 0s at the end.
2. We can also convert repeating decimal fractions into common frcations and then do addition or subtraction. In this method more additon or subtraction operations will be necessary.

**Example 10.** Add $3.\dot{8}\dot{9}$, $2.1\dot{7}\dot{8}$ and $5.89\dot{7}9\dot{8}$.

**Solution:** We have 2 digits in nonrepeating part, and repeating parts have respectively 2, 2 and 3 for which LCM is 6. At first all these three reepating decimal fractions have been converted into similar repeating decimal fractions.

$$
\begin{array}{rl}
3.\dot{8}\dot{9} & = 3.89\dot{8}98\dot{9}8\dot{9} \\
2.1\dot{7}\dot{8} & = 2.17\dot{8}78\dot{7}8\dot{7} \\
5.89\dot{7}9\dot{8} & = 5.89\dot{7}987\dot{9}8 \\
\hline
 & 11.97576574 \quad [8+8+7+2=25, \text{ here 2 is 2 in hands} \\
 & \qquad +2 \quad \text{here 2 of 25 has been added}] \\
\hline
 & 11.97\dot{5}76\dot{5}7\dot{6}
\end{array}
$$

Desired result is $11.97\dot{5}76\dot{5}7\dot{6}$ or, $11.97\dot{5}7\dot{6}$

**Remarks:** In the result 576576 is repeating part. But if we assume 576 is the repeating part then there will be no change of value .

**Note:** In order to clarify addition at the rightmost digit we have solved the addition in a different way:

$$
\begin{array}{rl}
3.\dot{8}\dot{9} & = 3.89\dot{8}98\dot{9}8\dot{9}|89 \\
2.1\dot{7}\dot{8} & = 2.17\dot{8}78\dot{7}8\dot{7}|87 \\
5.89\dot{7}9\dot{8} & = 5.89\dot{7}987\dot{9}8|79 \\
\hline
 & 11.97\dot{5}76\dot{5}7\dot{6}|55
\end{array}
$$

Here after the end of repating part some more digits have been taken. These extra digits have been separated by a vertical line. The addition has been performed. From the addition of digits just after the vertical line 2 in hands has been added to the digits just before the vertical line. It may be noted here the digit on the right side of the vertical line is the same as the first digit of repeatation.

**Example 11.** Add $8.9\dot{4}7\dot{8}$, $2.346 \quad 4.\dot{7}\dot{1}$.

**Solution:** In order to make repeating decimal fractions similar we must have 3 nonrepeating digits and repeating part must have digits equal to LCM of 2 and 3, that is 6. Now decimal fractions will be added.

$$
\begin{array}{rl}
8.9\dot{4}7\dot{8} & = 8.947\dot{8}4784\dot{7} \\
2.346 & = 2.346\dot{0}000\dot{0}0 \\
4.\dot{7}\dot{1} & = 4.717\dot{1}71\dot{7}1\dot{7} \\
\hline
 & 16.011019564 \quad [8+0+1+1=10, \text{ Here 1 is 1 in hand} \\
 & \qquad +1 \quad \text{Here 1 of 10 has been added}] \\
\hline
 & 16.011\dot{0}1956\dot{5}
\end{array}
$$

The desired sum is $16.011\dot{0}1956\dot{5}$

**Work:** Add 1) $2.0\dot{9}\dot{7} \quad 5.12\dot{7}6\dot{8}$ 2) $1.34\dot{5}, \; 0.31\dot{5}7\dot{6} \quad 8.056\dot{7}\dot{8}$

**Example 12.** Subtract $5.246\dot{7}\dot{3}$ from $8.2\dot{4}\dot{3}$.

**Solution:** Here number of digits in nonrepeating part should be 2 and that in repeating part should be 6, LCM of 2 and 3. Now the numbers have been converted into similar numbers, and subtraction has been performed as follows:

$$\begin{array}{rl} 8.2\dot{4}\dot{3} & = 8.24\dot{3}43\dot{4}3\dot{4} \\ 5.24\dot{6}7\dot{3} & = 5.24\dot{6}73\dot{6}7\dot{3} \\ \hline & 2.99669761 \quad [\text{ subtracting 6 from 3 results in 1 in hand }] \\ & \quad -1 \\ \hline & 2.99\dot{6}69760\dot{} \end{array}$$

The desired result is $2.99\dot{6}69760\dot{}$.

**Remarks:** If the minuend starting at the repeating dot is smaller than the subtrahend we must subtract 1 from the rightmost digit.

**Note:** In the following is explained why 1 is subtracted from the rightmost digit.

$$\begin{array}{rl} 8.2\dot{4}\dot{3} & = 8.24\dot{3}43\dot{4}3\dot{4}|34 \\ 5.24\dot{6}7\dot{3} & = 5.24\dot{6}73\dot{6}7\dot{3}|67 \\ \hline & 2.99\dot{6}69760\dot{}|67 \end{array}$$

The desired result is $2.99\dot{6}69760\dot{}|67$. We have obtained the same results in both the cases.

**Example 13.** Subtract $16.\dot{4}3\dot{7}$ from $24.45\dot{6}4\dot{5}$.

**Solution:**

$$\begin{array}{rl} 24.45\dot{6}4\dot{5} & = 24.45\dot{6}4\dot{5} \\ 16.\dot{4}3\dot{7} & = 16.43\dot{7}4\dot{3} \\ \hline & 8.01902 \quad \text{[if 7 is subtracted from 6 we have 1 in hand]} \\ & \quad -1 \\ \hline & 8.01\dot{9}0\dot{1} \end{array}$$

The desired result is $8.01\dot{9}0\dot{1}$

**Note:** In the following is explained why 1 is subtracted from the rightmost digit.

$$\begin{array}{rl} 24.45\dot{6}4\dot{5} & = 24.45\dot{6}4\dot{5}|64 \\ 16.\dot{4}3\dot{7} & = 16.43\dot{7}4\dot{3}|74 \\ \hline & 8.01\dot{9}0\dot{1}|90 \end{array}$$

**Work:** Subtract 1) Subtract $10.418$ from $13.12\dot{7}8\dot{4}$

2) Subtract $9.12\dot{6}4\dot{5}$ from $23.03\dot{9}\dot{4}$

### Multiplication and division of repeating decimal fractions

If repeating decimal fractions are first converted into common fractions and then multiplication or division operations are performed we can convert the result into repeating decimal fractions. If finite decimal fractions and repeating decimal fractions are multiplied or divided then this rule should be followed. However, in case of division if both divisor and dividend are repeating decimal fractions then converting them into similar repeating decimal fractions will ease the operations.

**Example 14.** Multiply $4.\dot{3}$ with $5.\dot{7}$.

**Solution:**

$$4.\dot{3} = \frac{43-4}{9} = \frac{39}{9} = \frac{13}{3}$$

$$5.\dot{7} = \frac{57-5}{9} = \frac{52}{9}$$

$$\therefore 4.\dot{3} \times 5.\dot{7} = \frac{13}{3} \times \frac{52}{9} = \frac{676}{27} = 25.\dot{0}3\dot{7}$$

The desired result is $25.\dot{0}3\dot{7}$.

**Example 15.** Multiply $0.2\dot{8}$ with $42.\dot{1}\dot{8}$.

**Solution:**

$$0.2\dot{8} = \frac{28-2}{90} = \frac{26}{90} = \frac{13}{45}$$

$$42.\dot{1}\dot{8} = \frac{4218-42}{99} = \frac{4176}{99} = \frac{464}{11}$$

$$\therefore\ 0.2\dot{8} \times 42.\dot{1}\dot{8} = \frac{13}{45} \times \frac{464}{11} = \frac{6032}{495} = 12.1\dot{8}\dot{5}$$

The desired result is $12.1\dot{8}\dot{5}$.

**Example 16.** How much is $2.5 \times 4.3\dot{5} \times 1.2\dot{3}\dot{4}$?

**Solution:**

$$2.5 = \frac{25}{10} = \frac{5}{2}$$

$$4.3\dot{5} = \frac{435 - 43}{90} = \frac{392}{90}$$

$$1.2\dot{3}\dot{4} = \frac{1234 - 12}{990} = \frac{1222}{990} = \frac{611}{495}$$

$$\therefore 2.5 \times 4.3\dot{5} \times 1.2\dot{3}\dot{4} = \frac{5}{2} \times \frac{392}{90} \times \frac{611}{495} = \frac{119756}{8910} = 13.440628\ldots$$

The desired result is 13.440628. (approx.)

**Work:** 1) Multiply $1.1\dot{3}$ with 2.6. 2) How much is $0.\dot{2} \times 1.\dot{1}\dot{2} \times 0.0\dot{8}\dot{1}$?

**Example 17.** Divide $7.\dot{3}\dot{2}$ by $0.2\dot{7}$.

**Solution:**

$$7.\dot{3}\dot{2} = \frac{732 - 7}{99} = \frac{725}{99}$$

$$0.2\dot{7} = \frac{27 - 2}{90} = \frac{25}{90} = \frac{5}{18}$$

$$\therefore 7.\dot{3}\dot{2} \div 0.2\dot{7} = \frac{725}{99} \div \frac{5}{18} = \frac{725}{99} \times \frac{18}{5} = \frac{290}{11} = 26.\dot{3}\dot{6}$$

The desired result $26.\dot{3}\dot{6}$.

**Example 18.** Divide $2.\dot{2}7\dot{1}\dot{8}$ by $1.9\dot{1}\dot{2}$.

**Solution:**

$$2.\dot{2}7\dot{1}\dot{8} = \frac{22718 - 2}{9999} = \frac{22716}{9999}$$

$$1.9\dot{1}\dot{2} = \frac{1912 - 19}{990} = \frac{1893}{990}$$

$$\therefore 2.\dot{2}7\dot{1}\dot{8} \div 1.9\dot{1}\dot{2} = \frac{22716}{9999} \div \frac{1893}{990} = \frac{22716}{9999} \times \frac{990}{1893} = \frac{120}{101} = 1.\dot{1}88\dot{1}$$

The desired result is $1.\dot{1}88\dot{1}$.

**Example 19.** Divide 9.45 by $2.8\dot{6}\dot{3}$.

**Solution:**

$$9.45 = \frac{945}{100} \qquad 2.8\dot{6}\dot{3} = \frac{2863 - 28}{990} = \frac{2835}{990}$$

$$\therefore 9.45 \div 2.8\dot{6}\dot{3} = \frac{945}{100} \div \frac{2835}{990} = \frac{945}{100} \times \frac{990}{2835} = \frac{189 \times 99}{2 \times 2835} = \frac{33}{10} = 3.3$$

The desired result is 3.3

**Remarks:** Multiplication and division of repeating decimal fractions may not be repeating decimal fractions.

**Work:** 1) Divide $0.\dot{6}$ by $0.\dot{9}$. 2) Divide $0.7\dot{3}\dot{2}$ by $0.0\dot{2}\dot{7}$.

### Infinite decimal fractions

There are many decimal numbers that have infinitely many digits after decimal point. Moreover, a digit or a number of digits do not repeat as well. These are called infinite decimal fractions. For example, 5.134248513942301 . . . is an infinite decimal fraction. Square root of 2 is an infinite decimal fraction. Let us calculate square root of 2.

```
 1 ) 2 ( 1.4142135...
     1
   ---
24 ) 100
      96
     ---
281 )  400
        281
       ---
2824 ) 11900
       11296
       -----
28282 )  60400
         56564
         -----
282841 )  383600
          282841
          ------
2828423 ) 10075900
          8485269
          -------
28284265 ) 159063100
           141421325
           ---------
            17641775
```

In this way the procedure may continue indefinitely. So $\sqrt{2} = 1.4142135\ldots$ an infinite decimal fraction.

### Value and approximate values up to certain decimal point

It is not the same to calculate an infinite decimal fraction up to decimal places or approximate it to certain decimal places. For example, $5.4325893\ldots$ when calculated up to 4 decimal places will give 5.4325 whereas its approximate value up to 4 decimal places will give us 5.4326. However, these two values are same if we calculate up to 2 decimal places. We can do the same for finite decimal fractions as well.

**Remarks:** For finding values up to certain decimal places we must write down those digits exactly as they are. However, for finding approximation up to certain decimal places we need to check the next digit. If that digit is 5 or more, 1 should be added to the last position. Otherwise it will be kept in tact.

**Example 20.** Find square root of 13, and write down its approximate value up to 3 decimal places.

**Solution:**

```
3   ) 13      (  3.605551...
      9
      ---
66  )  400
      396
      ----
7205 )  40000
        36025
        -----
72105 )  397500
          360525
          ------
721105 )  3697500
            3605525
            -------
7211101 )    9197500
               7211101
               -------
               1986399
```

$\therefore$ the desired square root is $3.605551\ldots$, and desired approximation up to 3 decimal places is 3.606

**Example 21.** Calculate values and approximate values of $4.4623845\ldots$ up to $1, 2, 3, 4$ and 5 decimal places ?

**Solution:** For the decimal fraction 4.4623845...

Value up to 1 decimal place is 4.4 but its approximate value up to 1 decimal digit is 4.5

Value up to 2 decimal places is 4.46 and approximate value is 4.46

Value up to 3 decimal places is 4.462 and approximate value is 4.462

Value up to 4 decimal places is 4.4623 and approximate value is 4.4624

Value up to 5 decimal places is 4.462238 and approximate value is 4.46238

**Work:** Calculate square root of 29 to 2 decimal places, and approximate it to 2 decimal places.

# Exercises 1.1

1. Which numbers are irrational?

   1) $0.\dot{3}$ 2) $\sqrt{\frac{16}{9}}$ 3) $\sqrt[3]{\frac{8}{27}}$ 4) $\frac{5}{\sqrt{3}}$

2. If $a,\ b,\ c,\ d$ are four consecutive natural numbers which one of the following will be a whole squared?

   1) $abcd$ 2) $ab+cd$ 3) $abcd+1$ 4) $abcd-1$

3. How many primes are there from 1 to 10 ? 1) 3 2) 4 3) 5 4) 6

4. Which one is the set of all integers?

   1) $\{\ldots,-4,\ -2,\ 0,\ 2,\ 4,\ldots\}$ 2) $\{\ldots,-2,\ -1,\ 0,\ 1,\ 2,\ldots\}$
   3) $\{\ldots,-3,\ -1\ 0,\ 1,\ 3,\ldots\}$ 4) $\{0,\ 1\ 2,\ 3,\ 4\}$

5. In case of real numbers

   (*i*) Square of an odd integer is odd.

   (*ii*) Product of two even numbers is even.

   (*iii*) Square root of a number that is not whole squared is an irrational number.

   Which one is true?

1) $i$ and $ii$ 2) $i$ and $iii$ 3) $ii$ and$iii$ 4) $i$, $ii$ and $iii$

6. Product of three consecutive numbers will always be divisible by the which of the following numbers?
   1) 5 2) 6 3) 7 4) 11

7. If $a$ and $b$ are two consecutive even numbers then which of the following numbers is odd?
   1) $a^2$ 2) $b^2$ 3) $a^2+1$ 4) $b^2+2$

8. If $a$ and $b$ are two integers then what should be added to $a^2+b^2$ to obtain a whole squared ?
   1) $-ab$ 2) $ab$ 3) $2ab$ 4) $ab$

9. Prove that the following numbers are irrational. 1) $\sqrt{5}$ 2) $\sqrt{7}$ 3) $\sqrt{10}$

10. 1) Find two irrational numbers between 0.31 and 0.12.
    2) Find a rational and an irrational number between $\frac{1}{\sqrt{2}}$ and $\sqrt{2}$.

11. 1) Prove that square of an odd integer is an odd integer.
    2) Prove that product of any two consecutive even integers is divisible by 8.

12. Convert the following into repating decimal fractions:
    1) $\frac{1}{6}$ 2) $\frac{7}{11}$ 3) $3\frac{2}{9}$ 4) $3\frac{8}{15}$

13. Convert into simple fractions
    1) $0.\dot{2}$ 2) $0.\dot{3}\dot{5}$ 3) $0.1\dot{3}$ 4) $3.7\dot{8}$ 5) $6.2\dot{3}0\dot{9}$

14. Express the follwoing into similar repeating decimal fractions.
    1) $2.2\dot{3}$, $5.2\dot{3}\dot{5}$ 2) $7.2\dot{6}$, $4.23\dot{7}$
    3) $5.\dot{7}$, $8.\dot{3}\dot{4}$, $6.\dot{2}4\dot{5}$ 4) $12.32$, $2.1\dot{9}$, $4.32\dot{5}\dot{6}$

15. Add:
    1) $0.4\dot{5}+0.1\dot{3}\dot{4}$ 2) $2.0\dot{5}+8.0\dot{4}+7.018$ 3) $0.00\dot{6}+0.\dot{9}\dot{2}+0.\dot{1}3\dot{4}$

16. Subtract:
    1) $3.\dot{4}-2.1\dot{3}$ 2) $5.\dot{1}\dot{2}-3.4\dot{5}$
    3) $8.49-5.3\dot{5}\dot{6}$ 4) $19.34\dot{5}-13.2\dot{3}4\dot{9}$

17. Multiply:
    1) $0.\dot{3}\times 0.\dot{6}$ 2) $2.\dot{4}\times 0.\dot{8}\dot{1}$ 3) $0.6\dot{2}\times 0.\dot{3}$ 4) $42.\dot{1}\dot{8}\times 0.2\dot{8}$

18. Divide:
    1) $0.\dot{3} \div 0.\dot{6}$ 2) $0.3\dot{5} \div 1.\dot{7}$ 3) $2.3\dot{7} \div 0.4\dot{5}$ 4) $1.\dot{1}8\dot{5} \div 0.\dot{2}\dot{4}$

19. Wrtite down values and approximate values up to 4 decimal places :
    1) 12 2) $0.\dot{2}\dot{5}$ 3) $1.\dot{3}\dot{4}$ 4) $5.1\dot{3}0\dot{2}$

20. Write down which of the following numbers are rational and irrational:
    1) $0.\dot{4}$ 2) $\sqrt{9}$ 3) $\sqrt{11}$ 4) $\frac{\sqrt{6}}{3}$
    5) $\frac{\sqrt{8}}{\sqrt{7}}$ 6) $\frac{\sqrt{27}}{\sqrt{48}}$ 7) $\frac{\frac{2}{3}}{\frac{3}{7}}$ 8) $5.\dot{6}3\dot{9}$

21. Let $n = 2x - 1$, where $x \in N$. Prove that $n^2$ when divided by 8 gives 1 as remainder.

22. $\sqrt{5}$ and 4 are two real numbers.
    1) Specify which one is rational and which one is irrational.
    2) Find two irrational numbers between $\sqrt{5}$ and 4.
    3) Prove that $\sqrt{5}$ is an irrational number.

23. Simplify:
    1) $(0.\dot{3} \times 0.8\dot{3}) \div (0.5 \times 0.\dot{1}) + 0.3\dot{5} \div 0.0\dot{8}$
    2) $[(6.27 \times 0.5) \div \{(0.5 \times 0.75) \times 8.36\}] \div$
       $\{(0.25 \times 0.1) \times (0.75 \times 21.\dot{3}) \times 0.5\}$